

Defence presentation outline

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1. Content (1 min)

- (a) I'll walk you through back ground my thesis addresses (Intro), both of these chapters (1 and 2) will address this question, but I'll start with the background first and finish with a conclusion; will last 15-20 minutes.
- (b) We don't have a lot of info on growing season length and growth relationship; I'll add data to this question in two systems that I find interesting.

2. Introduction

- (a) ACC lengthens growing seasons
 - i. Happens through earlier bigger shifts in SOS, and smaller delayed EOS
 - ii. (Figure): conceptual figure simplified showing SOS and EOS shifts
- (b) GS are longer, but also warmer which should increase growth
 - i. Total annual GDD increases
 - ii. (Figure): conceptual figure with GDD accumulation curve
- (c) SOS + EOS shifts = more GDD, but shifts are assymetric (both in how many days, but also for warmth)
 - i. Impacts how much GDD is accumulated in the GS
 - ii. SOS shift more, but are cooler
 - iii. EOS shift less, but are warmer
 - iv. (Figure): conceptual figure with showing how much GDD in each season
- (d) Recent studies failed to find this relationship (growth X GSL and thermal season)
 - i. Important because temperate forest= 40% of global net carbon sink
 - ii. (Figure): TBD
- (e) Question: do trees grow more with longer and warmer seasons?
 - i. Objectives: address GS/growth relationship using two systems
 - ii. Chapter 1: juvenile trees, 4 species, 4 prov, leaf phenology and tree rings
 - iii. Chapter 2: mature trees, 11 species, leaf phenology and tree rings
 - iv. (Figure): simplified version of thesis overview figure

3. Chapter 1

- (a) Introduction:
 - i. Drought and competition could constrain growth
 - ii. Internal constraints (local adaptation?)
 - iii. Answer if juvenile trees grow more in a drought and competition limited environment
 - iv. (Figure):
- (b) Methods: common garden set up
 - i. Trees grown from seeds, 4 species, 4 provenances
 - ii. Phenology for 3 years
 - iii. Calendar and thermal season from leafout to budset
 - iv. Growth via tree rings
 - v. Bayes hierarchical model

- vi. (Figure): map and common garden photo
- (c) R/D: thermal > calendar (question reminder)
 - i. Calendar season increased growth only for 2 species. Weaker effect is what people found in the literature
 - ii. 3 species grew more with warmer thermal season
 - iii. Species grew more with warmer seasons than longer calendar season
 - iv. Because thermal is a better predictor of growth
 - v. Thermal season weights growth days while calendar assumes a stable growth rate throughout the season
 - vi. (Figure): 2 pannel mu plot for GDD and GSL
- (d) R/D: EOS > SOS;
 - i. EOS temperature higher temperature than SOS
 - ii. EOS delayed less, but contribution is higher because days are warmer
 - iii. Suggests looking more at EOS than SOS
 - iv. (Figure): 2 pannel mu plot for EOS and SOS AND conceptual figure showing GDD differences at EOS and SOS
- (e) R/D: No provenance effect;
 - i. In this small climatic gradient and range, no evidence of local adaptation on growth
 - ii. Transition from juvenile to mature trees
 - iii. (Figure): provenance mu plot

4. Chapter 2

- (a) Introduction: (we have more data (but same data of rings and pheno), so well be able to ask this additional question on previous year conditions with a perpective on life history strategies could inform tree growth responses
 - i. Fast vs slower growing species
 - ii. Diffuse vs ring porous species
 - iii. Carbon allocation to storage = blur annual growth response to thermal and calendar seasons
- (b) Methods: citizen science leaf phenology at the Arboretum
 - i. Growth via tree rings
 - ii. Leaf phenology
 - iii. 11 angiosperm deciduous species spanning different growth strategies and wood porousness
 - iv. Bayes hierarchical model
 - v. (Figure): Arboretum map AND core scans
- (c) R/D: weak response across species across current year predictors
 - i. Ball decreased growth: southernmost range reduction in future ACC
 - ii. Tiam: grew less with earlier SOS despite being fast growing
 - iii. No consistent trend across life history strategies or wood porousness
 - iv. (Figure): mu plots 4 facet
- (d) R/D: 2 species responded to previous year thermal season
 - i. Ball increase growth with previous yr condition
 - ii. Bnig grew more only when considering previous yr condition
 - iii. Maybe suggests importance of carbon allocation to storage for mature trees; blur responses
 - iv. (Figure): previous year conditions mu plot

5. Conclusion and future work

- (a) Chapter 1 and 2
 - i. Chapter 1: warmer seasons for juvenile trees = good for forest regeneration and range expansion
 - ii. Chapter 2: no effect of longer and warmer seasons: probably bad for forest carbon
 - iii. (Figure): simplified version of thesis overview figure